

Lab 3: Component Modelling & Architectural Pattern Selection

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Problem Statement: Digital Parking Reservation Gateway

Architecture Selection

We chose Client-Server Architecture for the Digital Parking Reservation Gateway.

Reasons for this choice:

- **Centralized Management:** Client-server architecture provides centralized management of parking reservations, availability, violations and fine information. This suits DPRG because students, security staff and parking administrators need to access common parking data.
- **Easy Data Consistency:** The server can maintain a central database for parking slots, reservations and violations. This helps keep parking availability and reservation information consistent for different users.

Security Advantage

Security can be handled centrally using role-based access control. This allows different access permissions for students, security staff and parking administrators.

Performance Benefit

The server can process parking availability and reservation requests centrally, making it easier to provide updated parking information and handle multiple user requests.